Human Spaceflight: Gaganyaan and Planetary Missions - Solved Practice Workbook

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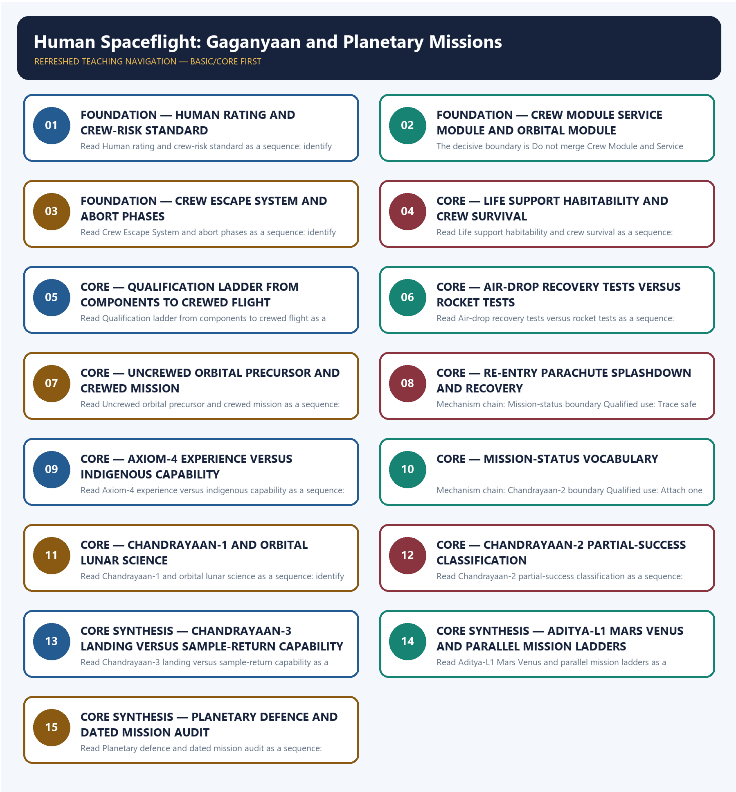

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Human-rating boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: A. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Human-rating boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: B. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Human-rating boundary without changing its institution, unit or status?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: C. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Human-rating boundary?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: D. Explanation: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Orbital-module boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: A. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Orbital-module boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: B. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Orbital-module boundary without changing its institution, unit or status?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: C. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Orbital-module boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: D. Explanation: Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Crew-escape boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: A. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Crew-escape boundary?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: B. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Crew-escape boundary without changing its institution, unit or status?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: C. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Crew-escape boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: D. Explanation: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Life-support boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: A. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Life-support boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: B. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a

satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Life-support boundary without changing its institution, unit or status?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: C. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Life-support boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.

Answer: D. Explanation: Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Qualification-ladder boundary?

A. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: A. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Qualification-ladder boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: B. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Qualification-ladder boundary without changing its institution, unit or status?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: C. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Qualification-ladder boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: D. Explanation: Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Air-drop-test boundary?

A. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: A. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Air-drop-test boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: B. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Air-drop-test boundary without changing its institution, unit or status?

A. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: C. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Air-drop-test boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.

Answer: D. Explanation: An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Uncrewed-crewed boundary?

A. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. B. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. C. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: A. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Uncrewed-crewed boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: B. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Uncrewed-crewed boundary without changing its institution, unit or status?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: C. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Uncrewed-crewed boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.

Answer: D. Explanation: A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an

achieved flight. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Recovery-chain boundary?

A. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: A. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Recovery-chain boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: B. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Recovery-chain boundary without changing its institution, unit or status?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: C. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Recovery-chain boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Answer: D. Explanation: A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Axiom-Gaganyaan boundary?

A. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: A. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Axiom-Gaganyaan boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: B. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Axiom-Gaganyaan boundary without changing its institution, unit or status?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: C. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Axiom-Gaganyaan boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.

Answer: D. Explanation: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital

module or recovery capability. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Mission-status boundary?

A. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: A. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Mission-status boundary?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: B. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Mission-status boundary without changing its institution, unit or status?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. C. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. D. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.

Answer: C. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Mission-status boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Answer: D. Explanation: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Chandrayaan-1 boundary?

A. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: A. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Chandrayaan-1 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: B. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Chandrayaan-1 boundary without changing its institution, unit or status?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: C. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Chandrayaan-1 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.

Answer: D. Explanation: Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Chandrayaan-2 boundary?

A. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: A. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Chandrayaan-2 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: B. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Chandrayaan-2 boundary without changing its institution, unit or status?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Chandrayaan-2 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.

Answer: D. Explanation: Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Chandrayaan-3 boundary?

A. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: A. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Chandrayaan-3 boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: B. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Chandrayaan-3 boundary without changing its institution, unit or status?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Chandrayaan-3 boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.

Answer: D. Explanation: Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Chandrayaan-4 boundary?

A. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: A. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Chandrayaan-4 boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: B. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Chandrayaan-4 boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: C. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Chandrayaan-4 boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.

Answer: D. Explanation: Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Aditya-L1 boundary?

A. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: A. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Aditya-L1 boundary?

A. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. B. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. C. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: B. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Aditya-L1 boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. D. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Answer: C. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Aditya-L1 boundary?

A. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.

Answer: D. Explanation: Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Lagrange-station-keeping boundary?

A. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: A. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Lagrange-station-keeping boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: B. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Lagrange-station-keeping boundary without changing its institution, unit or status?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Lagrange-station-keeping boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.

Answer: D. Explanation: A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Mars-Venus boundary?

A. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: A. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Mars-Venus boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: B. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Mars-Venus boundary without changing its institution, unit or status?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Mars-Venus boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. D. Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.

Answer: D. Explanation: Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Capability-ladder boundary?

A. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: A. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Capability-ladder boundary?

A. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. B. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: B. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Capability-ladder boundary without changing its institution, unit or status?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. C. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: C. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Capability-ladder boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Answer: D. Explanation: Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Planetary-defence boundary?

A. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. B. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: A. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Planetary-defence boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. C. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: B. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Planetary-defence boundary without changing its institution, unit or status?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. D. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.

Answer: C. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Planetary-defence boundary?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. D. Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.

Answer: D. Explanation: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile mission boundary?

A. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. B. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. C. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. D. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.

Answer: A. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile mission boundary?

A. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. B. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. C. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. D. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.

Answer: B. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile mission boundary without changing its institution, unit or status?

A. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. B. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. C. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. D. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Answer: C. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile mission boundary?

A. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. B. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. C. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. D. Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Answer: D. Explanation: Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2019 space-station, 2022 JWST, 2023 Chandrayaan-3 and 2024 asteroid Mains demands plus RTG, LISA and microgravity objective concepts here. No answer key, telescope specification or mission outcome is invented.

PYQ DEMAND CARD 1 - 2019 GS-III

Demand: Discuss India's space-station plan and its benefits for the space programme.

Status: Verified routed Mains demand.

Model solution: Human-rating boundary: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Life-support boundary:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Qualification-ladder boundary:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Uncrewed-crewed boundary:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Recovery-chain boundary:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Axiom-Gaganyaan boundary:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2019 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Human-rating boundary: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Life-support boundary:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Qualification-ladder boundary:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Uncrewed-crewed boundary:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Recovery-chain boundary:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Axiom-Gaganyaan boundary:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss India's space-station plan and its benefits for the space programme. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory

limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Human-rating boundary: Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Life-support boundary:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Qualification-ladder boundary:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Uncrewed-crewed boundary:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Recovery-chain boundary:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Axiom-Gaganyaan boundary:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2019 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 2 - 2022 GS-III

Demand: Discuss the features, goals and human benefits of the James Webb Space Telescope.

Status: Verified routed Mains demand; this package carries the mission-class and scientific-observatory method, not invented telescope specifications.

Model solution: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2022 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss the features, goals and human benefits of the James Webb Space Telescope. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; this package carries the mission-class and scientific-observatory method, not invented telescope specifications. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2022 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 3 - 2023 GS-III

Demand: Explain Chandrayaan-3 mission tasks, subsystems and the Virtual Launch Control Centre.

Status: Verified routed Mains demand; only owner-supported lunar capability and subsystem distinctions are used.

Model solution: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Chandrayaan-3 boundary:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Chandrayaan-4 boundary:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 3 - 2023 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Chandrayaan-3 boundary:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Chandrayaan-4 boundary:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Explain Chandrayaan-3 mission tasks, subsystems and the Virtual Launch Control Centre. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; only owner-supported lunar capability and subsystem distinctions are used. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Mission-status boundary: Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Chandrayaan-3 boundary:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Chandrayaan-4 boundary:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2023 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 4 - 2024 GS-III

Demand: Discuss asteroids, extinction threat and prevention strategies.

Status: Verified routed Mains demand.

Model solution: Planetary-defence boundary: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 4 - 2024 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Planetary-defence boundary: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss asteroids, extinction threat and prevention strategies. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Planetary-defence boundary: Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 4 - 2024 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 5 - 2024-2025 Prelims GS-I

Demand: RTG and microgravity-mission concept family.

Status: Verified routed objective concepts; official keys are not reproduced or inferred.

Model solution: Axiom-Gaganyaan boundary: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Mission-status boundary:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 5 - 2024-2025 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Axiom-Gaganyaan boundary: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Mission-status boundary:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: RTG and microgravity-mission concept family. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed objective concepts; official keys are not reproduced or inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Axiom-Gaganyaan boundary: Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Mission-status boundary:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Capability-ladder boundary:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Volatile mission boundary:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 5 - 2024-2025 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why human rating is more than an ordinary launcher success record. Answer in about 150 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:**

Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain why human rating is more than an ordinary launcher success record. Answer in about...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain why human rating is more than an ordinary launcher success record. Answer in about...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish Axiom-4 experience from indigenous Gaganyaan capability. Answer in about 150 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.
- Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.
- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish Axiom-4 experience from indigenous Gaganyaan capability. Answer in about 150...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability;

an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish Axiom-4 experience from indigenous Gaganyaan capability. Answer in about 150...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse the Gaganyaan qualification ladder before crewed flight. Answer in about 250 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:**

Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element.
- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
- An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved.
- A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary.

Named evidence/example: The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **analyse** requires a direct position on "Analyse the Gaganyaan qualification ladder before crewed flight. Answer in about 250 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

7. **Claim and named evidence:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbital-module boundary. **Named evidence/example:** Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Air-drop-test boundary. **Named evidence/example:** An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Uncrewed-crewed boundary. **Named evidence/example:** A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Analyse the Gaganyaan qualification ladder before crewed flight. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain India's lunar capability progression without overstating sample-return readiness. Answer in about 250 words.

Model thesis: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing.
- Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure.
- Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return.
- Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.

Qualified conclusion: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain India's lunar capability progression without overstating sample-return readiness....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Chandrayaan-1 boundary. **Named evidence/example:** Chandrayaan-1 was an orbital lunar-science mission associated with evidence of lunar hydroxyl or water; its science result is distinct from a soft landing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-2 boundary. **Named evidence/example:** Chandrayaan-2 achieved lunar-orbiter operations while its lander did not complete soft landing; the mission must not be labelled either total success or total failure. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-3 boundary. **Named evidence/example:** Chandrayaan-3 demonstrated a successful soft landing and surface operations; that achievement does not by itself establish lunar ascent, docking or sample return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain India's lunar capability progression without overstating sample-return readiness....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's human-spaceflight and planetary roadmap through parallel capability ladders. Answer in about 300 words.

Model thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Aditya-L1 boundary. **Named evidence/example:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mars-Venus boundary. **Named evidence/example:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.
- Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability.
- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.
- Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material.
- Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence.
- Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.
- Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Axiom-Gaganyaan boundary. **Named evidence/example:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Chandrayaan-4 boundary. **Named evidence/example:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Aditya-L1 boundary. **Named evidence/example:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mars-Venus boundary. **Named evidence/example:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; 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Demand decoding: The directive **evaluate** requires a direct position on "Evaluate India's human-spaceflight and planetary roadmap through parallel capability ladders....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; 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Analytical body:

1. **Claim and named evidence:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Axiom-4 gave an Indian astronaut experience on an international ISS mission using foreign launch and spacecraft systems; it did not demonstrate India's indigenous crewed launch, orbital module or recovery capability. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Chandrayaan-4 is an approved lunar sample-return technology mission in the repository owner; approval and intended completion are not launch, landing or returned material. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

7. **Claim and named evidence:** Aditya-L1 is a solar observatory operating in a halo orbit around Sun-Earth L1; it neither landed on the Sun nor belongs to the lunar or planetary-landing sequence. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Mars Orbiter Mission is a concluded historic mission, whereas the Venus Orbiter Mission is an approved future comparative-planetology mission in the owner; legacy capability does not prove the future mission's outcome. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
10. **Claim and named evidence:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Human-rating boundary. **Named evidence/example:** Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; 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legacy capability does not prove the future mission's outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Evaluate India's human-spaceflight and planetary roadmap through parallel capability ladders....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a status-disciplined science and safety framework for crewed, planetary and asteroid missions. Answer in about 300 words.

Model thesis: Claim: Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:**

Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lagrange-station-keeping boundary. **Named evidence/example:** A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Planetary-defence boundary. **Named evidence/example:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission.
- Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload.
- Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions.
- A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.
- Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted.
- A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point.
- Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another.
- Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield.
- Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test.

Qualified conclusion: Claim: Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; 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a Lagrange-point mission is not a motionless spacecraft at a force-free point. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Planetary-defence boundary. **Named evidence/example:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on "Design a status-disciplined science and safety framework for crewed, planetary and asteroid...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lagrange-station-keeping boundary. **Named evidence/example:** A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Capability-ladder boundary. **Named evidence/example:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Planetary-defence boundary. **Named evidence/example:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile mission boundary. **Named evidence/example:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

5. **Claim and named evidence:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** A halo orbit around L1 provides sustained solar viewing but requires station-keeping and a propellant budget; a Lagrange-point mission is not a motionless spacecraft at a force-free point. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Human spaceflight, lunar orbit-landing-sample return, solar observation and planetary orbit missions follow different technical ladders; progress on one ladder cannot certify another. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
8. **Claim and named evidence:** Asteroid risk management links detection, orbit characterisation, warning time and possible deflection; a kinetic-impact demonstration such as DART is evidence of one technique, not a complete planetary shield. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** Crew, schedule, test completion, payload, mission finding, spacecraft health, target date and operational status require a dated official mission source; no later rung may be inferred from an earlier announcement or test. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Crew-escape boundary. **Named evidence/example:** The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Life-support boundary. **Named evidence/example:** Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Qualification-ladder boundary. **Named evidence/example:** Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Recovery-chain boundary. **Named evidence/example:** A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Mission-status boundary. **Named evidence/example:** Announced, Cabinet-approved, under development, launched, inserted, operational, completed, partially successful, failed and concluded describe different mission states and must never be substituted. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lagrange-station-keeping boundary.

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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Design a status-disciplined science and safety framework for crewed, planetary and asteroid...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.